FORMAL MODIFICATIONS ARE LOCALLY ALGEBRAIZABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-formal-modifications-locally-algebraic; the proof environment immediately following it is its own proof

Editorial note. The label lemma-formal-modifications-locally-algebraic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- stability of formal modifications under adic base change (lemma-base-change-formal-modification of the same chapter, delivered as its own card)
- the dilatation theorem of the same chapter (theorem-dilatations): a formal modification of the completion of a locally Noetherian formal algebraic space along the closed fibre algebraizes to a proper morphism

Source reference. Original reference 0GDK: sources/stacks-restricted/pinned-original-source.tex, source lines 6275–6275. Prerequisite pointer invoked at 7405; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDN: sources/stacks-restricted/pinned-original-source.tex, source lines 6366–6366. Prerequisite pointer invoked at 7426; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDU: sources/stacks-restricted/pinned-original-source.tex, source lines 7376–7376. Prerequisite pointer invoked at 7427; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $\mathfrak{X}' \to \mathfrak{X}$ be a formal modification (Definition 0GDK) of locally Noetherian formal algebraic spaces over S. Given*

(1) *any adic Noetherian topological ring A,*
(2) *any adic morphism $Spf(A) \longrightarrow \mathfrak{X}$*

there exists a proper morphism $X \to \mathrm{Spec}(A)$ of algebraic spaces and an isomorphism

$$Spf(A) \times_{\mathfrak{X}} \mathfrak{X}' \longrightarrow X_{/Z}$$

over $Spf(A)$ of the base change of $\mathfrak{X}$ with the formal completion of X along the "closed fibre" $Z = X \times_{\mathrm{Spec}(A)} Spf(A)_{red}$ of X over A.

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Proof. The morphism $\mathrm{Spf}(A) \times_{\mathfrak{X}} \mathfrak{X}' \to \mathrm{Spf}(A)$ is a formal modification by Lemma 0GDN. Hence this follows from Theorem 0GDU. □